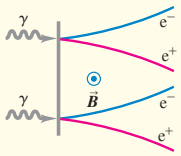
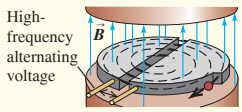
CHAPTER 44 SUMMARY

Fundamental particles: Each particle has an antiparticle; some particles are their own antiparticles. Particles can be created and destroyed, some of them (including electrons and positrons) only in pairs or in conjunction with other particles and antiparticles.

Particles serve as mediators for the fundamental interactions. The photon is the mediator of the electromagnetic interaction. Yukawa proposed the existence of mesons to mediate the nuclear interaction. Mediating particles that can exist only because of the uncertainty principle for energy are called virtual particles.



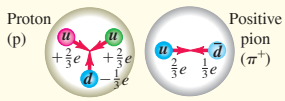
Particle accelerators and detectors: Cyclotrons, synchrotrons, and linear accelerators are used to accelerate charged particles to high energies for experiments with particle interactions. Only part of the beam energy is available to cause reactions with targets at rest. This problem is avoided in colliding-beam experiments. (See Examples 44.1–44.3.)



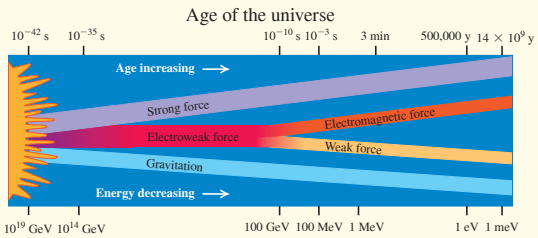
Particles and interactions: Four fundamental interactions are found in nature: the strong, electromagnetic, weak, and gravitational interactions. Particles can be described in terms of their interactions and of quantities that are conserved in all or some of the interactions.

Fermions have half-integer spins; bosons have integer spins. Leptons, which are fermions, have no strong interactions. Strongly interacting particles are called hadrons. They include mesons, which are always bosons, and baryons, which are always fermions. There are conservation laws for three different lepton numbers and for baryon number. Additional quantum numbers, including strangeness and charm, are conserved in some interactions. (See Examples 44.4–44.6.)

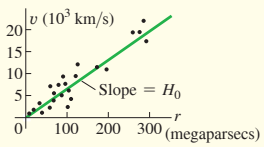
Quarks: Hadrons are composed of quarks. There are thought to be six types of quarks. The interaction between quarks is mediated by gluons. Quarks and gluons have an additional attribute called color. (See Example 44.7.)



Symmetry and the unification of interactions: Symmetry considerations play a central role in all fundamental-particle theories. The electromagnetic and weak interactions become unified at high energies into the electroweak interaction. In grand unified theories the strong interaction is also unified with these interactions, but at much higher energies.



The expanding universe and its composition: The Hubble law shows that galaxies are receding from each other and that the universe is expanding. Observations show that the rate of expansion is accelerating due to the presence of dark energy, which makes up 69.0% of the energy in the universe. Only 4.9% of the energy in the universe is in the form of conventional matter; the remaining 26.1% is dark matter, whose nature is poorly understood. (See Examples 44.8 and 44.9.)



The history of the universe: In the standard model of the universe, a Big Bang gave rise to the first fundamental particles. They eventually formed into the lightest atoms as the universe expanded and cooled. The cosmic background radiation is a relic of the time when these atoms formed. The heavier elements were manufactured much later by fusion reactions inside stars. (See Examples 44.10–44.12.)

Chapter 44 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLE 44.1 (Section 44.2) before attempting these problems.

VP44.1.1 In a cyclotron the maximum radius for the path of protons is 0.800 m. The magnetic field is 0.600 T. Find (a) the frequency of the alternating voltage applied to the dees, (b) the maximum proton kinetic energy in MeV, and (c) the maximum proton speed in terms of c.

VP44.1.2 The world's largest cyclotron, located at the TRIUMF laboratory in Vancouver, Canada, has a magnetic field of 0.460 T. It accelerates negative hydrogen ions with charge $-e$ and mass 1.67×10^{-27} kg. Find (a) the radius of the path of the ions when they have speed 6.00×10^6 m/s and (b) the angular frequency of the ions' motion.

VP44.1.3 The first cyclotron (in 1930) had a diameter of 11.4 cm and was used to accelerate negative hydrogen ions (charge $-e$, mass 1.67×10^{-27} kg) to kinetic energies of 80.0 keV. Find (a) the magnetic field used and (b) the frequency of the alternating voltage applied to the dees.

VP44.1.4 A cyclotron is used to accelerate alpha particles (charge $+2e$, mass 6.64×10^{-27} kg). The frequency of the alternating voltage applied to the dees is 5.80 MHz. Find (a) the magnetic field used and (b) the kinetic energy, in MeV, of the alpha particles when the radius of their circular path is 0.650 m.

Be sure to review EXAMPLES 44.2 and 44.3 (Section 44.2) before attempting these problems.

VP44.3.1 A proton with kinetic energy K collides with a proton at rest. Both protons remain after the collision along with an eta-prime (η') particle that has rest energy 958 MeV. In order for this reaction to happen, what must be (a) the minimum available energy in the center-of-momentum system and (b) the minimum value of K?

VP44.3.2 A proton with kinetic energy K collides with a proton at rest. The particles that remain after the collision are one proton and a delta-plus (Δ^+) that has rest energy 1232 MeV. In order for this reaction to happen, what must be (a) the minimum available energy in the center-of-momentum system and (b) the minimum value of K?

VP44.3.3 A negative pion (π^-) that has rest energy 140 MeV and kinetic energy K collides with a proton at rest. What remains after the collision is a delta-zero (Δ^0) that has rest energy 1232 MeV. In order for this reaction to happen, what must be (a) the minimum available energy in the center-of-momentum system and (b) the minimum value of K?

VP44.3.4 The Large Hadron Collider gives each proton a total energy of 7 TeV $= 7.00 \times 10^3$ GeV. (a) If a proton that has this energy collides with a second proton that is stationary, what is the available energy? (b) After the collision, both protons remain as well as a neutral particle X. What is the maximum mass of X?

Be sure to review EXAMPLES 44.8 and 44.9 (Section 44.6) before attempting these problems.

VP44.9.1 The red spectral line of hydrogen has a wavelength of 656.3 nm when the hydrogen is at rest. This line is observed to have a wavelength of 725 nm in the spectrum of a galaxy. Find (a) the redshift z and (b) the recession speed of the galaxy in terms of c.

VP44.9.2 A prominent absorption line in the spectrum of the sun (see Fig. 39.9) and many galaxies is due to singly ionized calcium. In the rest frame, this line has an ultraviolet wavelength of 396.9 nm. If a galaxy is receding from us at $0.0711c$, find (a) the wavelength at which we'll observe this line and (b) the redshift z.

VP44.9.3 Find the distances to (a) a galaxy with a recession speed of $0.0992c$ and (b) a galaxy with a recession speed of $0.0711c$. Give your answers in pc.

VP44.9.4 The spectrum of a Type Ia supernova (an exploding star so luminous that it can be seen even in distant galaxies) has an emission line due to singly ionized silicon. If astronomers observe this emission line at 615.0 nm in the laboratory but 666 nm in the spectrum of a Type Ia supernova in a distant galaxy, find (a) the recession speed of the galaxy in m/s and (b) the distance to the galaxy in pc.

BRIDGING PROBLEM Hyperons, Pions, and the Expanding Universe

(a) A Λ^0 hyperon decays into a neutron and a neutral pion (π^0). Find the kinetic energies of the decay products and the fraction of the kinetic energy that is carried off by each particle. (b) A π^0 is at rest in the galaxy shown in Fig. 44.20. If a physicist on earth detects one of the two photons emitted in the decay of this π^0, what is the energy of this detected photon?

SOLUTION GUIDE

IDENTIFY and SET UP

1. Which quantities are conserved in the Λ^0 decay? In the π^0 decay?
2. The universe expanded during the time that the photon traveled from the cluster to earth. How does this affect the wavelength and energy of the photon that the physicist detects?
3. List the unknown quantities for each part of the problem and identify the target variables.
4. Select the equations that will allow you to solve for the target variables.

EXECUTE

5. Write the conservation equations for the decay of the Λ^0. [*Hint:* It's useful to write the energy E of a particle in terms of its momentum p and mass m with $E = (p^2c^2 + m^2c^4)^{1/2}$.]
6. Solve the conservation equations for the energy of one of the decay products. (*Hint:* Rearrange the energy conservation equation so that one of the $(p^2c^2 + m^2c^4)^{1/2}$ terms is on one side of the equation. Then square both sides.) Then use $K = E - mc^2$.
7. Find the fraction of the total kinetic energy that goes into the neutron and into the pion.
8. Write the conservation equations for the decay of the π^0 at rest and find the energy of each emitted photon. By what factor does the wavelength of this photon change as it travels from the galaxy to earth? By what factor does the photon *energy* change? (*Hint:* See Fig. 44.20.)

EVALUATE

9. Which of the Λ^0 decay products should have the greater kinetic energy? Should the detected π^0 decay photon have more or less energy than when it was emitted?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q44.1 Is it possible that some parts of the universe contain antimatter whose atoms have nuclei made of antiprotons and antineutrons, surrounded by positrons? How could we detect this condition without actually going there? Can we detect these antiatoms by identifying the light they emit as composed of antiphotons? Explain. What problems might arise if we actually *did* go there?

Q44.2 Given the Heisenberg uncertainty principle, is it possible to create particle–antiparticle pairs that exist for extremely short periods of time before annihilating? Does this mean that empty space is really empty?

Q44.3 When they were first discovered during the 1930s and 1940s, there was confusion as to the identities of pions and muons. What are the similarities and most significant differences?

Q44.4 The gravitational force between two electrons is weaker than the electric force by the order of 10^{-40}. Yet the gravitational interactions of matter were observed and analyzed long before electrical interactions were understood. Why?

Q44.5 When a π^0 decays to two photons, what happens to the quarks of which it was made?

Q44.6 Why can't an electron decay to two photons? To two neutrinos?

Q44.7 According to the standard model of the fundamental particles, what are the similarities between baryons and leptons? What are the most important differences?

Q44.8 According to the standard model of the fundamental particles, what are the similarities between quarks and leptons? What are the most important differences?

Q44.9 The quark content of the neutron is ***udd***. (a) What is the quark content of the antineutron? Explain your reasoning. (b) Is the neutron its own antiparticle? Why or why not? (c) The quark content of the ψ is $c\bar{c}$. Is the ψ its own antiparticle? Explain your reasoning.

Q44.10 Does the universe have a center? Explain.

Q44.11 Does it make sense to ask, "If the universe is expanding, what is it expanding into?"

Q44.12 Assume that the universe has an edge. Placing yourself at that edge in a thought experiment, explain why this assumption violates the cosmological principle.

Q44.13 Explain why the cosmological principle requires that H_0 must have the same value everywhere in space, but does not require that it be constant in time.

EXERCISES

Section 44.1 Fundamental Particles—A History

44.1 • The starship *Enterprise*, of television and movie fame, is powered by combining matter and antimatter. If the entire 400-kg antimatter fuel supply of the *Enterprise* combines with matter, how much energy is released? How does this compare to the EU's yearly energy use, which is roughly 6.9×10^{19} J?

44.2 •• **CP** Two equal-energy photons collide head-on and annihilate each other, producing a $\mu^+\mu^-$ pair. The muon mass is given in terms of the electron mass in Section 44.1. (a) Calculate the maximum wavelength of the photons for this to occur. If the photons have this wavelength, describe the motion of the μ^+ and μ^- immediately after they are produced. (b) If the wavelength of each photon is half the value calculated in part (a), what is the speed of each muon after they have moved apart? Use correct relativistic expressions for momentum and energy.

44.3 •• A positive pion at rest decays into a positive muon and a neutrino. (a) Approximately how much energy is released in the decay? (Assume the neutrino has zero rest mass. Use the muon and pion masses given in terms of the electron mass in Section 44.1.) (b) Why can't a positive muon decay into a positive pion?

44.4 • A proton and an antiproton annihilate, producing two photons. Find the energy, frequency, and wavelength of each photon (a) if the p and $\bar{\text{p}}$ are initially at rest and (b) if the p and $\bar{\text{p}}$ collide head-on, each with an initial kinetic energy of 840 MeV.

44.5 •• **CP** For the nuclear reaction given in Eq. (44.2) assume that the initial kinetic energy and momentum of the reacting particles are negligible. Calculate the speed of the α particle immediately after it leaves the reaction region.

44.6 •• Estimate the range of the force mediated by an ω^0 meson that has mass 783 MeV/c^2.

Section 44.2 Particle Accelerators and Detectors

44.7 • In a collision experiment, a proton at rest is struck by an antiproton. (a) What is the minimum kinetic energy of the antiproton if the available energy is 2.00 TeV? (b) If a colliding beam is used instead of a stationary target, what minimum kinetic energy for each beam is required for an available energy of 2.00 TeV?

44.8 • An electron with a total energy of 14.7 GeV collides with a stationary positron. (a) What is the available energy? (b) If the electron and positron are accelerated in a collider, what total energy corresponds to the same available energy as in part (a)?

44.9 • Deuterons in a cyclotron travel in a circle with radius 30.3 cm just before emerging from the dees. The frequency of the applied alternating voltage is 8.30 MHz. Find (a) the magnetic field and (b) the kinetic energy and speed of the deuterons upon emergence.

44.10 • Bubble-chamber photographs taken with a 0.40 T magnetic field show at one point a positron moving perpendicular to the field in a 15 cm circle. What is the magnitude of the positron's momentum at that point?

44.11 • (a) A high-energy beam of alpha particles collides with a stationary helium gas target. What must the total energy of a beam particle be if the available energy in the collision is 16.2 GeV? (b) If the alpha particles instead interact in a colliding-beam experiment, what must the energy of each beam be to produce the same available energy?

44.12 • The first cyclotron used a 1.3 T magnetic field and had an 11 cm radius. (a) Find the maximum kinetic energy of protons accelerated by this cyclotron. Is it accurate to use nonrelativistic expressions in your calculation? (b) When the cyclotron was accelerating protons, at what frequency was it operated?

44.13 •• (a) What is the speed of a proton that has total energy 1050 GeV? (b) What is the angular frequency ω of a proton with the speed calculated in part (a) in a magnetic field of 3.50 T? Use both the nonrelativistic Eq. (44.7) and the correct relativistic expression, and compare the results.

44.14 •• Calculate the minimum beam energy in a proton-proton collider to initiate the $\text{p} + \text{p} \rightarrow \text{p} + \text{p} + \eta^0$ reaction. The rest energy of the η^0 is 547.9 MeV (see Table 44.3).

44.15 •• You work for a start-up company that is planning to use antiproton annihilation to produce radioactive isotopes for medical applications. One way to produce antiprotons is by the reaction $p + p \rightarrow p + p + p + \bar{p}$ in proton-proton collisions. (a) You first consider a colliding-beam experiment in which the two proton beams have equal kinetic energies. To produce an antiproton via this reaction, what is the required minimum kinetic energy of the protons in each beam? (b) You then consider the collision of a proton beam with a stationary proton target. For this experiment, what is the required minimum kinetic energy of the protons in the beam?

Section 44.3 Particles and Interactions

44.16 • In about 8% of Ω^- decays, the decay products are a Ξ^- and a π^0. (a) What is the energy released in this decay? (b) In the decay, what is the change in baryon number, and what is the change in strangeness? Your results should show that this decay is allowed for the weak nuclear interaction but not for the strong interaction.

44.17 • A K^+ meson at rest decays into two π mesons. (a) What are the allowed combinations of π^0, π^+, and π^- as decay products? (b) Find the total kinetic energy of the π mesons.

44.18 • How much energy is released when a μ^- muon at rest decays into an electron and two neutrinos? Neglect the small masses of the neutrinos.

44.19 • What is the mass (in kg) of the Z^0? What is the ratio of the mass of the Z^0 to the mass of the proton?

44.20 • The discovery of the Ω^- particle helped confirm Gell-Mann's eightfold way. If an Ω^- decays into a Λ^0 and a K^-, what is the total kinetic energy of the decay products?

44.21 • If a Σ^+ at rest decays into a proton and a π^0, what is the total kinetic energy of the decay products?

44.22 • Which of the following reactions obey the conservation of baryon number? (a) $p + p \rightarrow p + e^+$; (b) $p + n \rightarrow 2e^+ + e^-$; (c) $p \rightarrow n + e^- + \bar{\nu}_e$; (d) $p + \bar{p} \rightarrow 2\gamma$.

44.23 • In which of the following decays are the three lepton numbers conserved? In each case, explain your reasoning. (a) $\mu^- \rightarrow e^- + \nu_e + \bar{\nu}_\mu$; (b) $\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau$; (c) $\pi^+ \rightarrow e^+ + \gamma$; (d) $n \rightarrow p + e^- + \bar{\nu}_e$.

44.24 • In which of the following reactions or decays is strangeness conserved? In each case, explain your reasoning. (a) $K^+ \rightarrow \mu^+ + \nu_\mu$; (b) $n + K^+ \rightarrow p + \pi^0$; (c) $K^+ + K^- \rightarrow \pi^0 + \pi^0$; (d) $p + K^- \rightarrow \Lambda^0 + \pi^0$.

Section 44.4 Quarks and Gluons

44.25 • Determine the electric charge, baryon number, strangeness quantum number, and charm quantum number for the following quark combinations: (a) $\boldsymbol{uds}$; (b) $\boldsymbol{c\bar{u}}$; (c) $\boldsymbol{ddd}$; and (d) $\boldsymbol{d\bar{c}}$. Explain your reasoning.

44.26 •• Determine the electric charge, baryon number, strangeness quantum number, and charm quantum number for the following quark combinations: (a) $\boldsymbol{uus}$, (b) $\boldsymbol{c\bar{s}}$, (c) $\boldsymbol{\bar{d}\bar{d}\bar{u}}$, and (d) $\boldsymbol{\bar{c}b}$.

44.27 • Given that each particle contains only combinations of $\boldsymbol{u}, \boldsymbol{d}, \boldsymbol{s}, \boldsymbol{\bar{u}}, \boldsymbol{\bar{d}}$, and $\bar{s}$, use the method of Example 44.7 to deduce the quark content of (a) a particle with charge $+e$, baryon number 0, and strangeness $+1$; (b) a particle with charge $+e$, baryon number -1, and strangeness $+1$; (c) a particle with charge 0, baryon number $+1$, and strangeness -2.

44.28 • What is the total kinetic energy of the decay products when an upsilon particle at rest decays to $\tau^+ + \tau^-$?

Section 44.5 The Standard Model and Beyond

44.29 • Section 44.5 states that current experiments show that the mass of the Higgs boson is about 125 GeV/c^2. What is the ratio of the mass of the Higgs boson to the mass of a proton?

Section 44.6 The Expanding Universe

44.30 • The spectrum of the sodium atom is detected in the light from a distant galaxy. (a) If the 590.0 nm line is redshifted to 659 nm, at what speed is the galaxy receding from the earth? (b) Use the Hubble law to calculate the distance of the galaxy from the earth.

44.31 • A galaxy in the constellation Pisces is 5203 Mly from the earth. (a) Use the Hubble law to calculate the speed at which this galaxy is receding from earth. (b) What redshifted ratio λ_0/λ_S is expected for light from this galaxy?

44.32 • **Redshift.** The definition of the redshift z is given in Example 44.8. (a) Show that Eq. (44.13) can be written as $1 + z = ([1 + \beta]/[1 - \beta])^{1/2}$, where $\beta = v/c$. (b) The observed redshift for a certain galaxy is $z = 0.450$. Find the speed of the galaxy relative to the earth; assume the redshift is described by Eq. (44.14). (c) Use the Hubble law to find the distance of this galaxy from the earth.

Section 44.7 The Beginning of Time

44.33 • Calculate the reaction energy Q (in MeV) for the reaction $e^- + p \rightarrow n + \nu_e$. Is this reaction endoergic or exoergic?

44.34 • Calculate the energy (in MeV) released in the triple-alpha process $3\,{}^4He \rightarrow {}^{12}C$.

44.35 • Calculate the reaction energy Q (in MeV) for the nucleosynthesis reaction

$$ {}^{12}_{6}C + {}^{4}_{2}He \rightarrow {}^{16}_{8}O $$

Is this reaction endoergic or exoergic?

PROBLEMS

44.36 •• The strong nuclear force can be crudely modeled as a Hooke's-law spring force, increasing linearly with the quark separation distance. The energy stored in this "spring" corresponds to the energy content of the gluon field. In this picture, as quarks are further separated, increasing energy is stored between them. At a critical separation distance, the energy converts to matter, and a new quark–antiquark pair is generated, guaranteeing that there can never be a free quark. (a) A proton has a diameter of about 1.5 fm. Estimate the repulsive Coulomb force between two up quarks separated by 0.5 fm. (b) Model the strong force as $F_s = ks$, where s is the distance between two quarks. If this force balances the electrostatic repulsion between two up quarks when $s = 0.5$ fm, what is the effective spring constant k, in SI units? (c) Convert k into units of MeV/fm^2. (d) How much energy is stored in the gluon field when $s = 0.5$ fm? The mass of an up quark is thought to be about 2.3 MeV/c^2. (e) How much energy is needed to produce an up quark and an antiup quark? (f) How far would two up quarks need to be separated so that the gluon energy $\frac{1}{2}ks^2$ matches the rest energy of an up–antiup quark pair?

44.37 •• CP BIO **Radiation Therapy with π^- Mesons.** Beams of π^- mesons are used in radiation therapy for certain cancers. The energy comes from the complete decay of the π^- to *stable* particles. (a) Write out the complete decay of a π^- meson to stable particles. What are these particles? (b) How much energy is released from the complete decay of a single π^- meson to stable particles? (You can ignore the very small masses of the neutrinos.) (c) How many π^- mesons need to decay to give a dose of 50.0 Gy to 10.0 g of tissue? (d) What would be the equivalent dose in part (c) in Sv and in rem? Consult Table 43.3 and use the largest appropriate RBE for the particles involved in this decay.

44.38 •• A proton and an antiproton collide head-on with equal kinetic energies. Two γ rays with wavelengths of 0.900 fm are produced. Calculate the kinetic energy of the incident proton.

44.39 •• The 7 TeV proton bunches that circulate in opposite directions around the 27 km ring of the Large Hadron Collider smash together at beam crossings every 25 ns. (a) How many bunches meet every second? (b) Each bunch has 115 billion protons. Of these, typically 20 collide during each crossing. Estimate the fraction of protons that collide per crossing. (c) Estimate how many collisions take place each second. (d) The bunches are 30 cm long and are squeezed to a diameter of 20 μm. Estimate the density of protons in a bunch, in units of protons/mm^3. (e) Estimate the density of hadrons in ordinary matter. (*Hint:* Divide your mass, which is mostly due to hadrons, by the mass of a proton to get the number of hadrons in your body. Then divide by the estimated volume.)

44.40 •• Calculate the threshold kinetic energy for the reaction $\pi^- + p \rightarrow \Sigma^0 + K^0$ if a π^- beam is incident on a stationary proton target. The K^0 has a mass of 497.7 MeV/c^2.

44.41 • Each of the following reactions is missing a single particle. Calculate the baryon number, charge, strangeness, and the three lepton numbers (where appropriate) of the missing particle, and from this identify the particle. (a) $p + p \rightarrow p + \Lambda^0 + ?$; (b) $K^- + n \rightarrow \Lambda^0 + ?$; (c) $p + \bar{p} \rightarrow n + ?$; (d) $\bar{\nu}_\mu + p \rightarrow n + ?$

44.42 •• An η^0 meson at rest decays into three π mesons. (a) What are the allowed combinations of π^0, π^+, and π^- as decay products? (b) Find the total kinetic energy of the π mesons.

44.43 • The ϕ meson has mass 1019.4 MeV/c^2 and a measured energy width of 4.8 MeV/c^2. Using the uncertainty principle, estimate the lifetime of the ϕ meson.

44.44 • Estimate the energy width (energy uncertainty) of the ψ if its mean lifetime is 7.6×10^{-21} s. What fraction is this of its rest energy?

44.45 •• CP BIO One proposed proton decay is $p^+ \rightarrow e^+ + \pi^0$, which violates both baryon and lepton number conservation, so the proton lifetime is expected to be very long. Suppose the proton half-life were 1.0×10^{18} y. (a) Calculate the energy deposited per kilogram of body tissue (in rad) due to the decay of the protons in your body in one year. Model your body as consisting entirely of water. Only the two protons in the hydrogen atoms in each H_2O molecule would decay in the manner shown; do you see why? Assume that the π^0 decays to two γ rays, that the positron annihilates with an electron, and that all the energy produced in the primary decay and these secondary decays remains in your body. (b) Calculate the equivalent dose (in rem) assuming an RBE of 1.0 for all the radiation products, and compare with the 0.1 rem due to the natural background and the 5.0 rem guideline for industrial workers. Based on your calculation, can the proton lifetime be as short as 1.0×10^{18} y?

44.46 •• A ϕ meson (see Problem 44.43) at rest decays via $\phi \rightarrow K^+ + K^-$. It has strangeness 0. (a) Find the kinetic energy of the K^+ meson. (Assume that the two decay products share kinetic energy equally, since their masses are equal.) (b) Suggest a reason the decay $\phi \rightarrow K^+ + K^- + \pi^0$ has not been observed. (c) Suggest reasons the decays $\phi \rightarrow K^+ + \pi^-$ and $\phi \rightarrow K^+ + \mu^-$ have not been observed.

44.47 ••• CP About 10,000 cosmic-ray protons, each with hundreds of MeV of energy, impinge on each square meter of our upper atmosphere each second. They collide with atmospheric nitrogen and oxygen to produce secondary showers of newly created particles, including many muons. Muons have a mass of 105.7 MeV/c^2 and an average lifetime of 2.197 μs. Consider a secondary cosmic muon produced at an altitude of 15.00 km aimed directly downward with an energy of 6.000 GeV. With such high energy, the muon can travel a great distance into the earth without slowing down. (a) Determine the speed of this muon. (b) How far would this muon travel in one lifetime if there were no relativistic effects? (c) In the frame of the muon, what distance separates its creation position from the earth's surface? (d) If one lifetime passes in the frame of the muon, how much time passes in the frame of the earth? (e) How far does a muon travel in this time as seen from the earth? (f) Does the muon survive its trip to the surface? How far will it penetrate the earth in its lifetime?

44.48 ••• CP A Ξ^- particle at rest decays to a Λ^0 and a π^-. (a) Find the total kinetic energy of the decay products. (b) What fraction of the energy is carried off by each particle? (Use relativistic expressions for momentum and energy.)

44.49 ••• CP A Σ^- particle moving in the $+x$-direction with kinetic energy 180 MeV decays into a π^- and a neutron. The π^- moves in the $+y$-direction. What is the kinetic energy of the neutron, and what is the direction of its velocity? Use relativistic expressions for energy and momentum.

44.50 ••• CP The K^0 meson has rest energy 497.7 MeV. A K^0 meson moving in the $+x$-direction with kinetic energy 225 MeV decays into a π^+ and a π^-, which move off at equal angles above and below the $+x$-axis. Calculate the kinetic energy of the π^+ and the angle it makes with the $+x$-axis. Use relativistic expressions for energy and momentum.

44.51 •• DATA While tuning up a medical cyclotron for use in isotope production, you obtain the data given in the table.

B (T)	0.10	0.20	0.30	0.40
K_{max} (MeV)	0.068	0.270	0.608	1.080

B is the uniform magnetic field in the cyclotron, and K_{max} is the maximum kinetic energy of the particle being accelerated, which is a proton. The radius R of the proton path at maximum kinetic energy has the same value for each magnetic-field value. (a) Compare the kinetic energy values in the table to the rest energy mc^2 of a proton. Is it necessary to use relativistic expressions in your analysis? Explain. (b) Graph your data as K_{max} versus B^2. Use the slope of the best-fit straight line to your data to find R. (c) What is the maximum kinetic energy for a 0.25 T magnetic field? (d) What is the angular frequency ω of the proton when $B = 0.40$ T?

44.52 •• DATA The decay products from the decay of short-lived unstable particles can provide evidence that these particles have been produced in a collision experiment. As an initial step in designing an experiment to detect short-lived hadrons, you make a literature study of their decays. Table 44.3 gives experimental data for the mass and typical decay modes of the particles Σ^-, Ξ^0, Δ^{++}, and Ω^-. (a) Which of these four particles has the largest mass? The smallest? (b) By the decay modes shown in the table, for which of these particles do the decay products have the greatest total kinetic energy? The least?

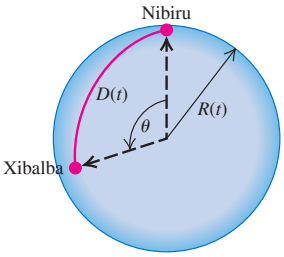
44.53 •• DATA You have entered a graduate program in particle physics and are learning about the use of symmetry. You begin by repeating the analysis that led to the prediction of the Ω^- particle. Nine of the spin-$\frac{3}{2}$ baryons are four Δ particles, each with mass 1232 MeV/c^2, strangeness 0, and charges $+2e$, $+e$, 0, and $-e$; three Σ^* particles, each with mass 1385 MeV/c^2, strangeness -1, and charges $+e$, 0, and $-e$; and two Ξ^* particles, each with mass 1530 MeV/c^2, strangeness -2, and charges 0 and $-e$. (a) Place these particles on a plot of S versus Q. Deduce the Q and S values of the tenth spin-$\frac{3}{2}$ baryon, the Ω^- particle, and place it on your diagram. Also label the particles with their masses. The mass of the Ω^- is 1672 MeV/c^2; is this value consistent with your diagram? (b) Deduce the three-quark combinations (of $\boldsymbol{u}$, $\boldsymbol{d}$, and $\boldsymbol{s}$) that make up each of these ten particles. Redraw the plot of S versus Q from part (a) with each particle labeled by its quark content. What regularities do you see?

44.54 ••• CP A positive kaon K^+ which is a bound state of an up quark and an antidown quark, can decay into an antimuon and a muon neutrino via a weak interaction process mediated by a W^+ boson. We can determine how much of the kaon's energy each of the daughter particles obtains using relativistic kinematics. (a) In the rest frame of the kaon, the speed of the antimuon is v and its Lorentz factor is $\gamma = (1 - v^2/c^2)^{-1/2}$. Using the notation E_ν for the energy of the neutrino, M_K for the mass of the kaon, and M_μ for the mass of the antimuon, write the relativistic expression for conservation of energy. Treat the neutrino as massless. (b) Write the relativistic equation for conservation of momentum. (c) Solve these two equations to determine v as a function of the mass ratio $\sigma = M_K/M_\mu$. (d) Using the masses M_K and M_μ found in Tables 44.3 and 44.2, respectively, determine the value of σ. (e) What is the energy of $E_\mu = \gamma M_\mu c^2$ of the antimuon? (f) What is the energy of the neutrino? (g) Do these energies add up to the rest energy of the kaon?

CHALLENGE PROBLEMS

44.55 ••• CALC Consider the following hypothetical universe: The Nibiruvians and the Xibalbans are minuscule "flat" societies located at an angular separation of $\theta = 60°$ on the surface of a two-dimensional spherical universe with radius R, as shown in **Fig. P44.55.** (a) What is the distance D between the two societies in terms of R and θ? (b) If R is increasing in time, what is the speed $V = dD/dt$ at which the civilizations are separating, as a function of R, dR/dt, and D? (c) The separation velocity and the distance D are related by $V = BD$. Determine the "Bubble parameter" B in terms of $R(t)$. Note that B is not constant. (Similarly, the Hubble "constant" H_0 is presumed to be changing with time.) (d) The radius of this universe was $R_0 = 500.0$ m upon its creation at time $t = 0$, and it has been increasing at a constant rate of $1.00\,\mu$m/s. What was the Bubble parameter exactly four years after creation? (e) What is the distance between Nibiru and Xibalba four years after creation? (f) At what speed are they separating at that time? (g) At that time, the Nibiruvians transmit isotropic ripple waves with their "light speed" of $c = 6.35\,\mu$m/s and wavelength 1.00 nm. How long does it take these waves to reach Xibalba? (*Hint:* In time dt the waves travel an angular distance $d\theta = c\,dt/R(t)$. Integrate to obtain an expression for the total angular distance traveled as a function of time. Solve for the time needed to travel a given angular distance.) (h) At what wavelength will the Xibalbans observe these waves? (*Hint:* Use Eq. (44.16).)

Figure P44.55



44.56 ••• CP Consider a collision in which a stationary particle with mass M is bombarded by a particle with mass m, speed v_0, and total energy (including rest energy) E_m. (a) Use the Lorentz transformation to write the velocities v_m and v_M of particles m and M in terms of the speed v_{cm} of the center of momentum. (b) Use the fact that the total momentum in the center-of-momentum frame is zero to obtain an expression for v_{cm} in terms of m, M, and v_0. (c) Combine the results of parts (a) and (b) to obtain Eq. (44.9) for the total energy in the center-of-momentum frame.

MCAT-STYLE PASSAGE PROBLEMS

BIO **Looking Under the Hood of PET.** In the imaging method called positron emission tomography (PET), a patient is injected with molecules containing atoms that have nuclei with an excess of protons. As they decay into neutrons, these protons emit positrons. An emitted positron travels a short distance and slows to near-zero velocity; when it encounters an electron, they may annihilate each other and emit two photons in opposite directions. The patient is enclosed in a circular array of detectors, with the tissue to be imaged centered in the array. If two photons of the proper energy strike two detectors simultaneously (within 10 ns), we can conclude that the photons were produced by positron–electron annihilation somewhere along a line connecting the detectors. By observing many such simultaneous events, we can create a map of the distribution of positron-emitting atoms in the tissue. However, photons can be absorbed or scattered as they pass through tissue. The number of photons remaining after they travel a distance x through tissue is given by $N = N_0 e^{-\mu x}$, where N_0 is the initial number of photons and μ is the attenuation coefficient, which is approximately 0.1 cm^{-1} for photons of this energy. The index of refraction of biological tissue for x rays is 1.

44.57 What is the energy of each photon produced by positron–electron annihilation? (a) $\frac{1}{2}m_e v^2$, where v is the speed of the emitted positron; (b) $m_e v^2$; (c) $\frac{1}{2}m_e c^2$; (d) $m_e c^2$.

44.58 Suppose that positron–electron annihilations occur on the line 3 cm from the center of the line connecting two detectors. Will the resultant photons be counted as having arrived at these detectors simultaneously? (a) No, because the time difference between their arrivals is 100 ms; (b) no, because the time difference is 200 ms; (c) yes, because the time difference is 0.1 ns; (d) yes, because the time difference is 0.2 ns.

44.59 If the annihilation photons come from a part of the body that is separated from the detector by 20 cm of tissue, what percentage of the photons that originally traveled toward the detector remains after they have passed through the tissue? (a) 1.4%; (b) 8.6%; (c) 14%; (d) 86%.

ANSWERS

Chapter Opening Question ?

(v) Only 4.9% of the mass and energy of the universe is in the form of "normal" matter. Of the rest, 26.1% is poorly understood dark matter and 69.0% is even more mysterious dark energy.

Key Example VARIATION Problems

VP44.1.1 (a) 9.15 MHz (b) 11.0 MeV (c) $0.153c$
VP44.1.2 (a) 0.136 m (b) 4.41×10^7 rad/s
VP44.1.3 (a) 0.717 T (b) 10.9 MHz
VP44.1.4 (a) 0.756 T (b) 11.6 MeV
VP44.3.1 (a) 2834 MeV (b) 2.41×10^3 MeV $= 2.41$ GeV
VP44.3.2 (a) 2170 MeV (b) 634 MeV
VP44.3.3 (a) 1232 MeV (b) 190 MeV
VP44.3.4 (a) 115 GeV (b) 113 GeV/c^2
VP44.9.1 (a) 0.105 (b) $0.0992c$
VP44.9.2 (a) 426 nm (b) 0.0738
VP44.9.3 (a) 4.4×10^8 pc (b) 3.1×10^8 pc
VP44.9.4 (a) 2.38×10^7 m/s (b) 3.5×10^8 pc

Bridging Problem

(a) neutron: 5.78 MeV (0.140 of total);
pion: 35.62 MeV (0.860 of total)
(b) 8.1 MeV